# Unsupervised Domain Adaptation for Remote Sensing Image Semantic Segmentation Using Region and Category Adaptive Domain Discriminator

Xiaoshu Chen, Shaoming Pan, and Yanwen Chong

***Abstract*— By reason of factors such as terrains, weather conditions, sensor imaging methods, and cultural and economic development, there is a large shift between the remote sensing imagery collected from different geographic locations and different sensors, which makes the state-of-the-art semantic segmentation models trained on source domain (an image set gathered from specific geographic locations and sensors) difficult to generalize to target domain (another image set collected from other geographic locations and sensors). Currently, unsupervised domain adaptation (UDA) using adversarial training, whose purpose is to align the marginal distribution in the output space between the source and target domains, is the most explored and practical approach to address this issue. However, this global alignment approach does not take into account diversities of different regions in a specific image nor the category-level distribution, which leads to the consequence that some regions and categories which are already well aligned between the source and target domains may be incorrectly remapped. Therefore, we propose a region and category adaptive domain discriminator (RCA-DD), aiming to emphasize the differences in regions and categories during the process of alignment. Specifically, on the one hand, we propose an entropy-based regional attention module (ERAM) in domain discriminator to emphasize the importance of difficult-to-align regions. On the other hand, we propose a class-clear module (CCM) to update only the distribution of existing categories in one iteration without affecting all categories. Finally, a lot of experiments are introduced to indicate that the proposed method can obtain better results when compared with other state-of-the-art UDA methods using adversarial training.**

***Index Terms*— Remote sensing image processing, semantic segmentation, unsupervised domain adaptation (UDA).**

## I. Introduction

SEMANTIC segmentation of remote sensing images, as a task of assigning class labels of ground to all pixels in an image, has made great progress with the help of the development of convolutional neural networks [1], [2], [4]. However, since the intense variations in scenes and style of remote sensing imagery resulting from terrains, weather conditions, sensor imaging methods, and cultural and economic development, the remote sensing community is facing a problem in this task that even if the same categories exist in the source domain and the target domain, a well-trained model trained on a source domain is unable to generalize to target domain which is collected in different geographic locations, weather conditions, or sensors.

A simple solution is to retrain this model on the target domain. But since abundant available remote sensing images are unlabeled, it is not practical to retrain for each target domain. Therefore, the unsupervised domain adaptation (UDA) that can generate high-quality segmentations on target domain without any annotations is proposed. Among the methods of UDA [5], [6], [7], [8], [9], [10], adversarial training is the most explored and practical direction. Adversarial training mainly includes two components: 1) semantic segmentation model to output prediction of two domains and 2) domain discriminator to identify which domain the output comes from. Under the supervision of adversarial loss, the segmentation model will produce an output to fool the domain discriminator so that the predicted label distribution of the source image and the target image can be aligned in the output space. Once the distributions is aligned, the segmentation model can be generalized to both domains.

However, on the one hand, traditional adversarial training may map the originally aligned distribution of category to the wrong spatial position (negative transfer). Negative transfer results from a major limitation: as long as the segmentation model aligns the global marginal distribution of the predicted label for two domains in the output space, the domain discriminator cannot distinguish which domain the output comes from [10]. As shown in Fig. 1(a) (left), to align global marginal distribution to source domain, the distributions of target domain only needs to be transferred as a whole in output space without considering the independence of local categories, which give rise to that the well aligned distributions of class A are remapping into an error position during global movement.

On the other hand, according to [18], different regions of the same image in the target domain have different

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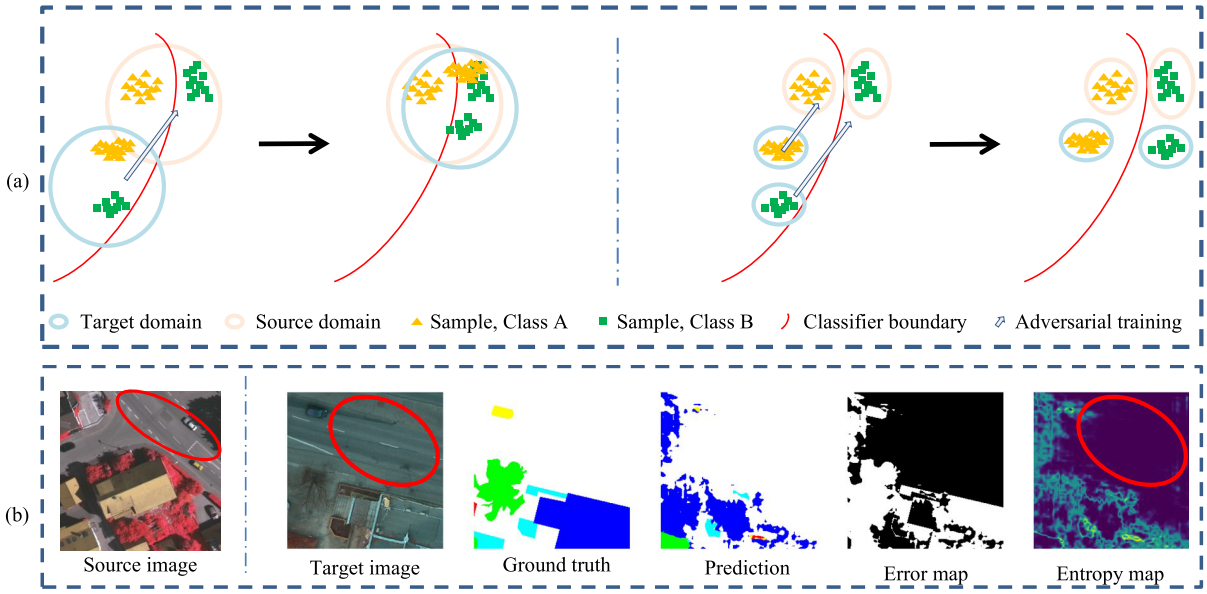



Fig. 1. Illustration of adversarial training and entropy map. (a) Comparison of traditional adversarial training methods with our category adaptive domain discriminator. Our method pursues the distribution alignment not from the global perspective but from the perspective of each category, which avoids negative transfer. (b) Comparison between entropy map and error map obtained by the model before adversarial training. The highlighted regions in the entropy map and the white regions in error maps indicate higher entropy and prediction error, respectively.

difficulty in distribution alignment. Thus, aligning the features of each region in the image in the same way will cause the distributions of the some regions that could have been aligned to deviate from the correct position in the output space during adversarial training. However, although Wang *et al.* [18] notices this problem and divides the regions in the image into stuff and things whose distribution will be aligned separately, this artificial division is not adaptive because there are regions in the interior of stuff and things where the difficulty of distribution alignment is different.

Therefore, we propose a region and category adaptive domain discriminator (RCA-DD) that includes entropy-based regional attention module (ERAM) and class-clear module (CCM) to solve the above problems. CCM uses a category aggregation strategy to separately calculate the adversarial loss for each category appearing in the current image so that the alignment of the categories is independent and does not interfere with each other. As shown in Fig. 1(a) (right), each class adaptively aligns their distribution in the output space in its own most suitable way and do not interfere with others during alignment. This independent alignment of category distribution in output space can avoid negative transfer. And ERAM regard the entropy map which can reflect the accurateness of prediction generated by model [9] as the attention map to force adversarial training to mainly transfer distributions of regions that are difficult to align and reduce the distribution transfer of regions that are easy to align. As shown in Fig. 1(b), the high entropy regions basically correspond to the inaccurate prediction regions. In other words, regions with high entropy correspond to regions in the image where features are difficult to align because most of the inaccurate predictions in the target domain are caused by the misalignment of feature distributions. The correspondence between the entropy map and the region alignment difficulty is more intuitively reflected in the red ellipse of Fig. 1(b). As shown by the red ellipse, regions in the target domain image that are relatively similar in texture and color to the source domain image have lower values in the entropy map while the rest regions whose distribution discrepancy are large have higher entropy values. Therefore, applying the entropy map as the attention map actually mines the prior information of the difficulty of aligning distribution for different regions, so that the discriminator can adaptively align the distribution of different region with different intensities.

The main contributions of this article are as follows.

1) *Class-Clear Module:* It makes the adversarial training not to make the alignment process a global movement, but a separate transfer for distribution of each class without interfering with each other.
2) *Entropy-Based Regional Attention Module:* It mines the prior information about the difficulty to align distribution of the regions contained in the entropy map so that the model can adaptively focus on transferring the regions where the distribution are difficult to align.
3) *Better Performance:* A large number of experiments show that our proposed method is better than other state-of-the-art UDA methods.

## II. Related Work

In this section, we introduce the unsupervised domain adaptive methods of semantic segmentation in the field of natural images and remote sensing images.

### A. Unsupervised Domain Adaptation of Natural Image

In general, the works on UDA of semantic segmentation in natural images are divided into three directions: self-supervised domain adaptation, image-to-image translation, and adversarial learning.

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*1) Self-Supervised Domain Adaptation:* The core idea of self-supervised learning is to train the model by generating pseudo-labels from the previous state of the model. Selective pseudo-labeling (SPL) [49] and robust Class-wise Matching (RCM) [48] generate pseudo label through mapping the images of the source and target domains into a unified feature space to achieve classwise feature alignment. However, the two have to treat each pixel as a separate representation when applied to the semantic segmentation, which ignores the crucial contextual relationship for semantic segmentation. Class-balanced self-training (CBST) is proposed to generate class-balanced pseudo labels from images that were assigned labels according to the last state of model [22]. To further improve the reliability of pseudo labels, spatial priors [20], [23], [24] is introduced. However, an inherent disadvantage of self-supervised learning is that there are usually contain noisy predictions in pseudo labels, which greatly limits the performance of domain adaptation. Although setting dynamic thresholds [20] and involving the uncertainty into the optimization objective [25] can alleviate this defect, this issue cannot be solved in essence.

*2) Image-to-Image Translation:* The image-to-image translation first transforms the color and texture of the source domain to be similar to the target domain and then train the segmentation model on the source domain [13], [14], [15], [16]. Since the visual gap between the source domain and the target domain is reduced, the model is easier to generalize to the target domain. Under the guidance of this methodology, CyclyGAN [13] is proposed to adapt cycle-consistent adversarial networks to learn a mapping from source domain to target domain. Fourier domain adaptation (FDA) [14] swaps the low-frequency spectrum of one with the other by Fourier transform and its inverse. But it should be noted that the translated images may generate a lot of noise when mapping, which causes that the annotations of the original source domain images and the translated source domain images do not match.

*3) Adversarial Learning:* The research direction of adversarial learning has gone through a long course of development. Yaroslav *et al.* [38] adapts adversarial learning for UDA of image classification. Borrowing from the structure of Yaroslav *et al.*, Hoffman *et al.* [5] applies the adversarial training for UDA on semantic segmentation for the first time. To take structural consistency across domain into count, Adapt-SegNet [7] uses adversarial training on output space and ADVENT [9] applies weighted self-information to further optimize the performance of distribution alignment on output space. But none of the above methods consider the independence between categories and the differences between regions in the image. To overcome these two issues, category-level adversaries network (CLAN) [10] is proposed to introduce a category-level adversarial network to ensure local semantic consistency during global alignment. Wang *et al.* [18] divides regions in an image into stuff and things, and then independently aligns the feature distribution of these two regions. However, Since CLAN does not introduce clear category information, the distribution alignment of different categories is still not independent and will affect each other, resulting in CLAN still not eliminating the impact of global alignment. And artificially dividing the picture into different regions does not adaptively select regions that are difficult to align in the image. Therefore, on the one hand, we introduced a CCM to independently align the distribution of each class. On the other hand, we proposed an ERAM to adaptively align distribution of different regions with different intensities.

### *B. Unsupervised Domain Adaptation of Remote Sensing Image*

In the field of UDA for remote sensing image semantic segmentation, how to reduce the distribution shift between the source domain and the target domain is a key issue [37]. There are many traditional unsupervised remote sensing image processing methods, such as histogram matching [27], kernel principal component analysis (KPCA) [29], graph matching [28], color constancy algorithms [30], and minimizing the statistical distance between the source domain and the target domain through the maximum mean discrepancy (MMD) [32], which can align the data distributions. Although the aforementioned method is concise, the performance of them is poor. Thus, centroid and covariance alignment [33], nonlinear feature normalization [34], and optimal transport [35] are proposed to further improve the performance of UDA for remote sensing image semantic segmentation.

Recently, with the popularity of deep learning methods of UDA for nature image semantic segmentation, some works begin to study how to draw lessons from these methods to UDA for remote sensing image. For example, since there are a lot of heterogeneous and complex structures in remote sensing image, ColorMapGAN [26] is proposed to perform image to image translation that is suitable for remote sensing images. As far as we know, there are few works that try adversarial training on UDA for remote sensing image semantic segmentation. Thus, our work fills this gap and greatly improved the performance of UDA for remote sensing image.

## III. METHOD

To construct our models, we add an additional network branch (domain discriminator) used for domain adaptation on the basis of existing semantic segmentation models. Fig. 2 illustrates our architectures.

To better understand, we first formulate some concepts. For an image set $\mathcal{X}_s \subset \mathbb{R}^{H\times W\times 3}$ of source domain samples, there are corresponding ground-truth $C$-class segmentation maps $\mathcal{Y}_s \subset (1, C)^{H\times W}$, where $H$ and $W$ represents height and width of image, respectively. Obviously, there is not a corresponding ground-truth set for the target image set $\mathcal{X}_t \subset \mathbb{R}^{H\times W\times 3}$. And we denote $x_s \subset \mathcal{X}_s$, $x_t \subset \mathcal{X}_t$, and $y_s \subset \mathcal{Y}_t$ as sample of corresponding set.

Let $F$ and $D$ denote segmentation network and domain discriminator, respectively. $F$ takes image $x$ as input and output $C$-dimensional prediction probability map $F(x, \theta_F) = P_x \subset \mathbb{R}^{H\times W\times C}$, where $\theta_F$ is parameters of $F$. $0 \leq [P_x]_c^n \leq 1$ represents probability that the pixel at $n$ space position in $c$ dimensional is classified to class $c$. $D$ processes $P_x$ and then generates $D_x \subset (0-1)^{H\times W}$ that indicates if the inputs

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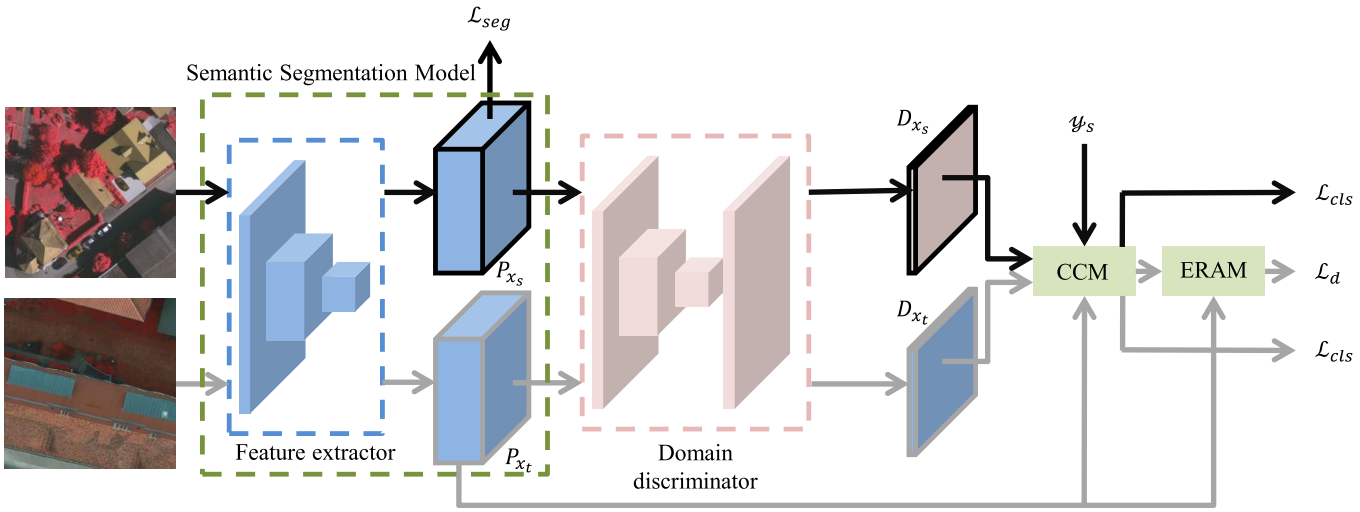



Fig. 2. Illustration of our model. The black and gray lines represent source domain flow and target domain flow.

correspond to source or target domain at any space position. $\theta_D$ denotes the parameters of $D$.

### A. Overview

In this part, we introduce a novel adversarial training framework which uses RCA-DD to minimizes the distribution shift between the source domain and target domain. RCA-DD consists of a traditional domain classifier $D$ followed by the CCM and the ERAM. In addition to RCA-DD, the adversarial training framework contains a segmentation network $F$. The segmentation network first is trained with segmentation loss $L_{\text{seg}}(P_{x_s}, y_s)$ that can be formulated as

$$L_{\text{seg}}\left(P_{x_s}, y_s\right) = \sum_{n=0}^{N}\sum_{c=0}^{C} -[y_s]_c^n \cdot \log\left[P_{x_s}\right]_c^n \tag{1}$$

where $N = H \times W$. As for the RCA-DD that is designed to distinguish between source domain and target domain predicted representations adaptively, to understand the role of the domain discriminator in general, we temporarily do not insert the CCM and ERAM, but directly take $D_{x_s}$ and $D_{x_t}$ generated by domain discriminator $D$ as the final output. On the one hand, to make domain discriminator have ability to distinguish the predicted representations generated by $F$, we train it with classification loss $L_{\text{cls}}$ that can be formulated as

$$\begin{aligned} L_{\text{cls}}\left(D_{x_s}, D_{x_t}\right) = &-\sum_{n=0}^{N} \log\left(\sigma\left(\left[D_{x_s}\right]^n\right)\right)/N \\ &-\sum_{n=0}^{N} \log\left(1-\sigma\left(\left[D_{x_t}\right]^n\right)\right)/N \end{aligned} \tag{2}$$

where $\sigma$ is the sigmoid function. Obviously, the domain discriminator is actually a binary classifier. The source domain is labeled as 1 and the target domain is 0.

On the other hand, to make the predicted representation from the source domain and the target domain indistinguishable so as to fool the domain discriminator, the segmentation network second has to be trained by the adversarial loss $L_d$ that is defined as follows:

$$L_d\left(D_{x_t}\right) = -\sum_{n=0}^{N} \log\left(\sigma\left(\left[D_{x_t}\right]^n\right)\right)/N. \tag{3}$$

In summary, the training objective of domain discriminator is

$$\min_{\theta_D} L_{\text{cls}}\left(D_{x_s}, D_{x_t}\right). \tag{4}$$

And combining (1) and (3), we derive the optimization objective of segmentation network

$$\min_{\theta_F}\left(L_{\text{seg}}\left(P_{x_s}, y_s\right) + \lambda_d L_d\left(D_{x_t}\right)\right) \tag{5}$$

with the weighting factor $\lambda_d$ for the adversarial object $L_d$. During training, we alternatively optimize $\theta_D$ and $\theta_F$ with objective functions in (4) and (5).

After inserting CCM and ERAM, traditional domain discriminator is transformed to RCA-DD but the overall process has not changed, the difference is that we apply CCM and ERAM to process $D_{x_s}$ and $D_{x_t}$ for the purpose of achieving a separate and adaptive transfer for each class and enforcing the segmentation model adaptively to focus on transferring the regions where the features are difficult to align. The detailed structure of CCM and ERAM can be seen in Fig. 3.

### B. Class-Clear Module

In (2) and (3), $L_{\text{cls}}$ and $L_d$ simply aggregate the loss at each spatial location $n$ in $D_{x_t}$ and $D_{x_t}$ by calculating their average, which essentially treats the distribution of the source domain and target domain as a whole without distinguishing the category of each spatial location, resulting in the global movement of the target domain distribution. In this way, the transfer of the distribution of each category may affect other categories and lead to negative transfer. For example, there are class A and class B but no class C in an image. Obviously, in an iteration of model training, we should only adjust the distribution of A and B without adjusting the distribution of C.

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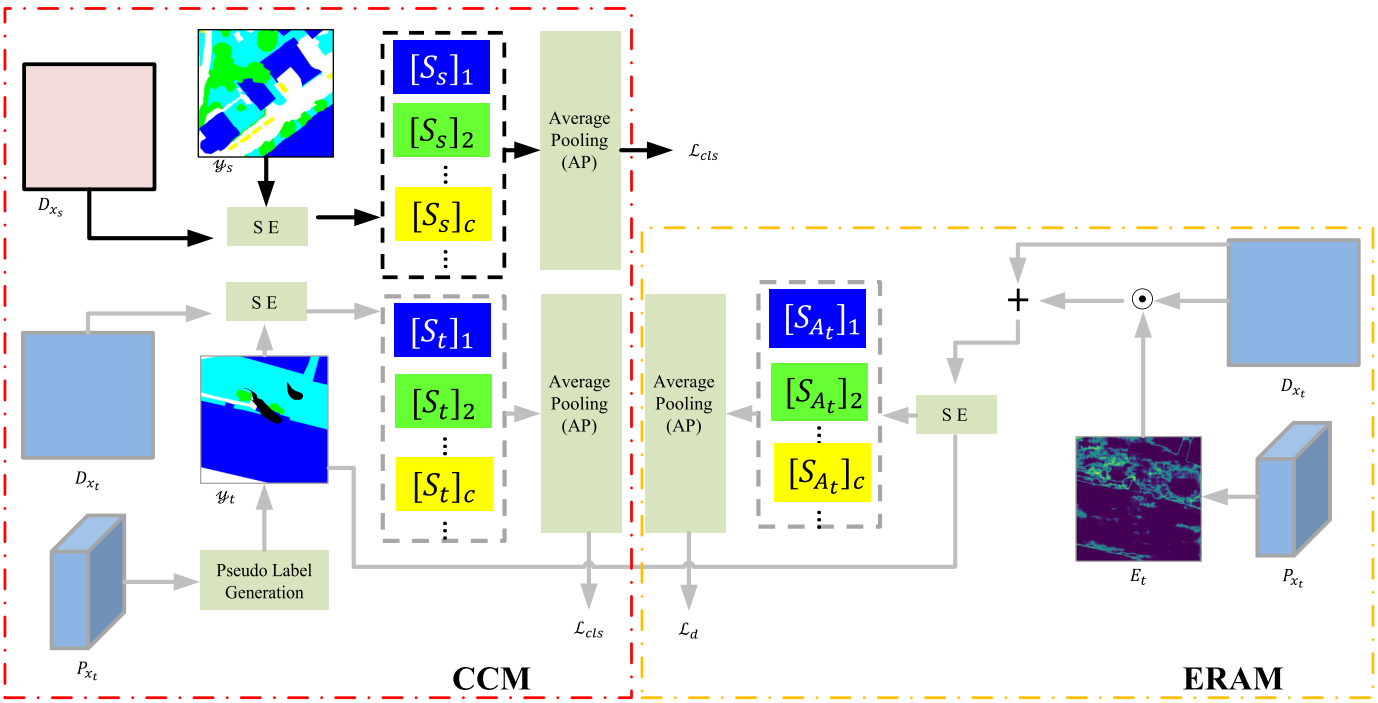



Fig. 3. Details of CCM and ERAM. The black and gray lines represent source domain flow and target domain flow. The legend of $y_s$ and $y_t$: white—impervious surfaces; blue—building; light blue—low vegetation; green—tree; and black—unlabeled.

In addition, the distribution of A and class B should be shifted according to their own circumstances. However, since $L_{\text{cls}}$ and $L_d$ treat the distribution of the entire domain as a whole, the distributions of A, B, and C will be transformed in the same way in this iteration. Thus, in CCM, we first clarify which category each spatial location belongs to.

For $D_{x_s}$, since the category at each spatial location $n$ has been labeled in the corresponding $y_s$, we can directly utilize $y_s$ to obtain category information. But for $D_{x_t}$, since there are not any labels, we perform a pseudo label generation strategy as below to obtain pseudo label $y_t$

$$[y_t]^n = \begin{cases} c & \left[P_{x_t}\right]_c^n \geq \varphi_c \text{ and } c = \underset{c'}{\operatorname{argmax}} \left[P_{x_t}\right]_{c'}^n \\ -1 & \text{otherwise} \end{cases} \tag{6}$$

where $\varphi_c$ denotes the probability threshold and -1 represents there is no reliable label at $n$. Once $y_t$ is obtained, we can use $y_t$ and $y_s$ to obtain the set $[S_t]_c = \{[D_{x_t}]_c^{i_1}, [D_{x_t}]_c^{i_2}, \ldots, [D_{x_t}]_c^{i_t}, \ldots\}$ and $[S_s]_c = \{[D_{x_s}]_c^{j_1}, [D_{x_s}]_c^{j_2}, \ldots, [D_{x_s}]_c^{j_s}, \ldots\}$, where $[D_{x_t}]_c^{i_t}$ and $[D_{x_s}]_c^{j_s}$ represent the value of $D_{x_t}$ and $D_{x_s}$ at the spatial position $i_t$ and $j_s$ that belong to class $c$ in $y_t$ and $y_s$, respectively. We denote this selection process as $[S]_c = \text{SE}(D_x, y, c)$.

And then, we revise (2) as follows:

$$L_{\text{cls}}(S_s, S_t) = -\sum_c \log(\sigma(\text{AP}([S_s]_c))) - \sum_c \log(1 - \sigma(AP([S_t]_c)) \tag{7}$$

where AP(·) refers to a category aggregation strategy that adopts average pooling for the corresponding category set to aggregate the category to a single overall representation. Specifically, AP(·) can be formulated as

$$\text{AP}([S]_c) = \left(\sum_{m=0}^{M} [D_x]_c^m\right) \Big/ M \tag{8}$$

where $M$ is the number of elements in the set $[S]_c$. And in the same way, (3) can be revised to

$$L_d(S_t) = -\sum_c \log(\sigma(\text{AP}([S_t]_c))). \tag{9}$$

Compared with the original global aggregation strategy that does not distinguish categories, the proposed category aggregation strategy explicitly tells the domain discriminator to determine which domain the input comes from based on the corresponding category features alone instead of directly judging from a global perspective.

In general, these revision is able to bring two benefit. First, each category is individually aggregated through average pooling, and then the alignment of distribution is promoted through minimizing $L_{\text{cls}}$ and $L_d$ that take the aggregated value as input, which makes the transfer of each category no longer uniform but adaptive and not interfere with each other. Second, the distribution of category not existing in current image will not be transferred because there is not an aggregation value for the category since the pseudo label generation strategy will not generate labels for nonexistent categories. The above two points are both effective in preventing negative transfer.

### C. Entropy-Based Regional Attention Module

Since different spatial locations are also not distinguished in (3), the distributions of different spatial regions in the target

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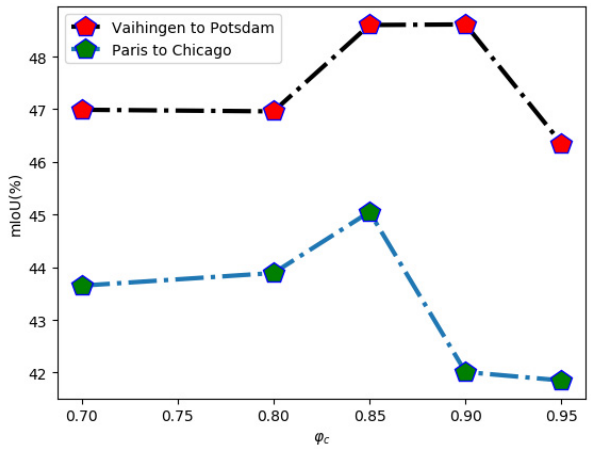



Fig. 4. Performance of the model under different $\varphi_c$.

domain are actually aligned to the source domain with the same intensity. However, due to different contents contained in different regions, the difficulty to transfer the distributions of different regions into a distribution similar to the source domain is different. Regions that difficult to transfer need more intensity while regions that are easy to transfer only need to slightly transfer.

Therefore, according to the analysis for Fig. 1(b) as mentioned above, to reflect the difficulty of different regions, we calculate Shannon Entropy

$$[E_t]^n = -\sum_{c=0}^{C} \left[P_{x_t}\right]_c^n \cdot \log\left(\left[P_{x_t}\right]_c^n\right) \tag{10}$$

to generate the entropy map $E_t$. The higher the value of $(E_t)^n$, the more difficult it is to align the distribution at $n$. So, we apply $E_t$ as the attention map to enforce model to focus on the regions that are difficult to transfer, which can be formulated as

$$\left[A_{x_t}\right]^n = \sigma\left([E_t]^n\right) \cdot \left[D_{x_t}\right]^n + \left[D_{x_t}\right]^n. \tag{11}$$

In (11), $\sigma([E_t]^n) \cdot [D_{x_t}]^n$ can be regarded as a weighting operation, which makes most use of entropy map to adjusts the corresponding weight at different positions adaptively. In addition, according to ResNet [39], an identity term $[D_{x_t}]^n$ is added at the end of (11) to maintain the stability of model training.

Then, taking $A_{x_t}$ and $y_t$ as the input, we perform $\mathrm{SE}(A_{x_t}, y_t, c)$ to get the set $[S_{A_t}]_c$. Therefore, (9) is transformed to

$$L_d\left(S_{A_t}\right) = -\sum_{c} \log\left(\sigma\left(\mathrm{AP}\left(\left[S_{A_t}\right]_c\right)\right)\right). \tag{12}$$

Since the regions are weighted in (11), the distribution of difficult-to-align regions has a greater impact on the revised loss function $L_d$, which forces the model to push these regions with greater intensity.

It should be noted that we only use the ERAM when fooling the domain discriminator and not when training the domain discriminator. This is because the ERAM is equivalent to telling the domain discriminator which regions are different between the source domain and the target domain, so that the domain discriminator itself does not need to make a judgment and therefore loses its ability to discriminate.

## IV. Experiments

### A. Experimental Detail

*1) Datasets:* Due to the influence of terrains, weather conditions, sensor imaging methods and cultural and economic development, and other factors, remote sensing images collected in different cities often have a large gap in data distribution. Therefore, we adopt the setting of adaptation from source city to the target city to test the performance of our method. Specifically, the following two sets of experiments are include.

*a) Vaihingen to Potsdam:* The ISPRS Vaihingen and ISPRS Potsdam datasets contain very high-resolution (VHR) remote sensing images from two German cities: Vaihingen and Potsdam, respectively. Their imaging methods and ground sampling distances are different. Among them, the Vaihingen dataset and Potsdam dataset collects 33 and 38 VHR images that are cropped from the larger VHR ortho-remote sensing images. And both of them contain six categories, including impervious surfaces, buildings, low vegetation, trees, cars, and background/clutter.

Due to the limitation of GPU, we used random cropping method to crop images from Vaihingen and Potsdam to a size of 512 × 512. After cropping, the Vaihingen and Potsdam datasets produced 3000 and 6000 images, respectively. The Vaihingen dataset is regarded as the source domain while Potsdam is the target domain. All the images in the source domain participate in the training of the model. And in the target domain, we randomly divide 30% of the images as the test set, and the remaining 70% as the training set.

*b) Paris to Chicago:* The CITY-OSM dataset [47] consists of aerial images from Berlin, Chicago, Paris, Potsdom, and Zurich of which pixelwise building, road, and background labels are provided. We chose the images from Paris and Chicago to verify our model since the images collected from these two cities are very different in vision. The Paris dataset and Chicago dataset include 725 and 457 VHR ortho-remote sensing, respectively.

In the same way, we crop images to a size of 512 × 512. After cropping, the Paris and Chicago datasets produced 33 807 and 20 994 images, respectively. The setting of the ratio of the training set to the test set is consistent with the experiment like Vaihingen to Potsdam.

*2) Network Architecture:* For semantic segmentation network $F$, deeplab v2 [1] is applied. Deeplab v2 consists of a feature extractor resnet101 [39] followed by an atrous spatial pyramid pooling (ASPP) whose sampling rates are fixed as 6, 12, 18, 24. Following [1], atrous convolution is used in the conv5 of resnet101 to obtain a larger receptive field. In addition, we perform multilevel outputs strategy [7] to further improve the performance of semantic segmentation network.

As for the domain discriminator $D$, deep convolutional generative adversarial networks (DCGAN) [8] is applied. In particular, the probability map $P_x$ is fed into four convolutional layers, each followed by a leaky-rectified linear unit (ReLU)

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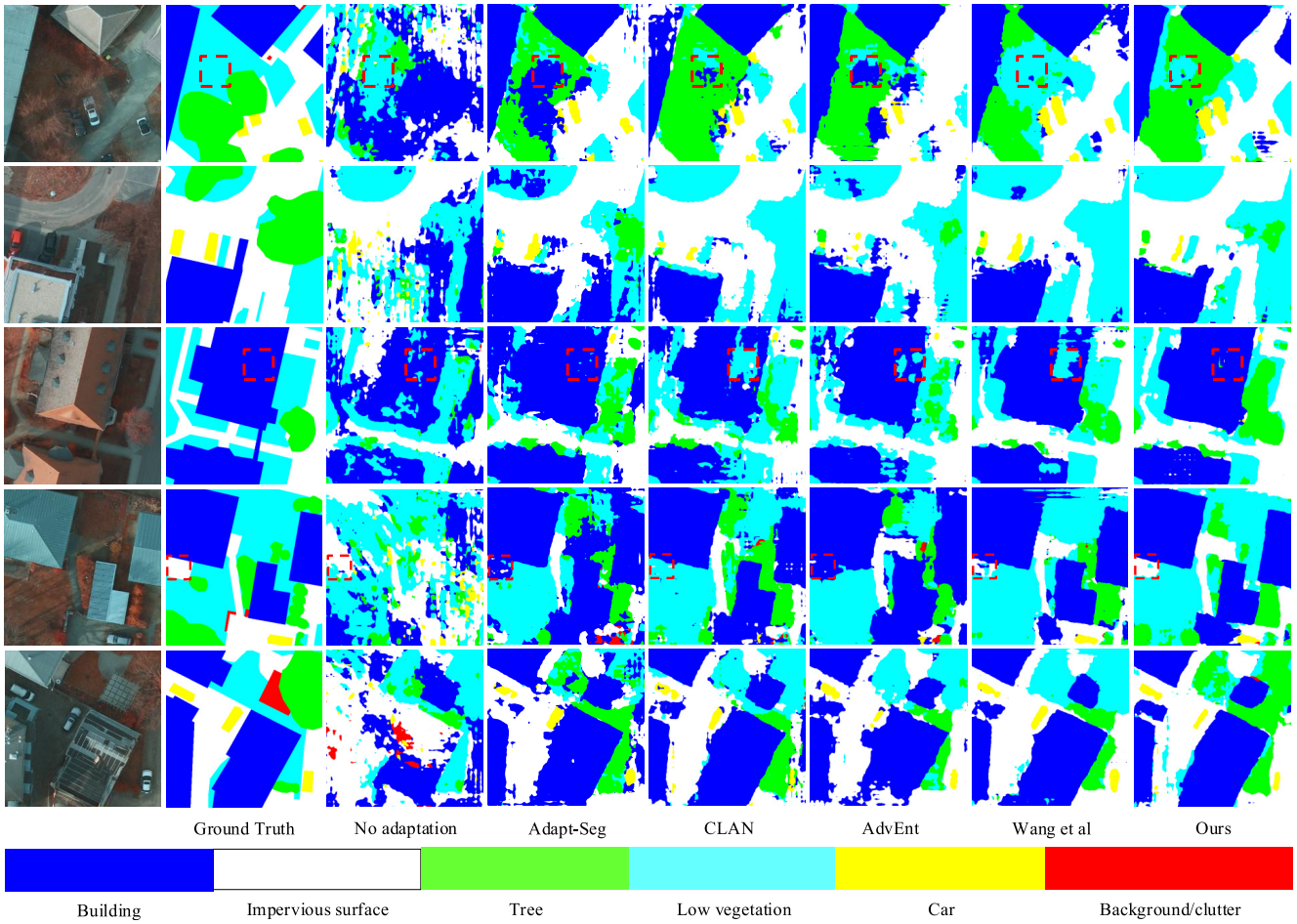



Fig. 5. Visualization of segmentation maps on Potsdam test set.

layer. Finally, a $1 \times 1$ convolutional layers is performed to produces classification outputs that indicate which domain are the inputs come from.

*3) Implementation Detail:* We illustrate the detailed process of model training in Algorithm 1. The entire model is implemented by PyTorch [40]. And we train the model on a single NVIDIA 1080TI GPU. Semantic segmentation network $F$ is trained with stochastic gradient descent (SGD) optimizer [41] with momentum 0.9, learning rate 0.00025, and weight decay 0.001. As for domain discriminator $D$, the optimizer we use is Adam optimizer [42] with learning rate 0.001. And the polynomial annealing procedure mentioned in [1] is applied to adjust the learning rate.

For $\varphi_c$, we test the performance of the model under different $\varphi_c$. The result is shown in Fig. 4. Based on it, when $\varphi_c$ is set to 0.85, the model can achieve the best result. As for the weighting factor $\lambda_d$, we fix it to 0.001 according to the setting of ADVENT [9].

When testing, we only need to use the segmentation network $F$ and not the discriminator $D$. The entire testing process is no different from conventional semantic segmentation model testing.

*4) Evaluation Metric:* We apply per-class Intersection over Union (IoU)

$$\text{IoU} = \frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}} \tag{13}$$

to evaluate the segmentation performance of each class, where TP, FP, and FN indicate true positive, false positive, and false negative counts, respectively. And we calculate the

**Algorithm 1** Training Process of the Proposed Method

**Input:** Training data $x_s$, $y_s$, $x_t$, training epochs $n_e$, learning rate of segmentation netwok $\eta_F$, and learning rate of discriminator $\eta_D$.

Initialize parameters $\theta_F$ of the segmentation network $F$ by the parameters pretrained on ImageNet.
Randomly initialize parameters $\theta_D$ of the discriminator $D$.
**for** epoch in $n_e$ **do**
  **while** iteration is effective **do**
    Generate $P_{x_s}$ by $F(x_s, \theta_F)$, and generate $P_{x_t}$ by $F(x_t, \theta_F)$
    Update $\theta_F$: $\theta_F \leftarrow l_{seg} - \eta_F \nabla_{\theta_F} l_{seg}$;
    Generate $D_{x_s}$ by $D(P_{x_s}, \theta_D)$ and generate $D_{x_t}$ by $D(P_{x_t}, \theta_D)$
    Process $D_{x_s}$ and $D_{x_t}$ by CCM and ERAM
    Update $\theta_D$: $\theta_D \leftarrow l_{cls} - \eta_D \nabla_{\theta_D} l_{cls}$
    Frozen $\theta_D$ and Update $\theta_F$: $\theta_F \leftarrow \lambda_d l_d - \eta_F \nabla_{\theta_F} \lambda_d l_d$
  **end while**
**end for**

**Output:** The trained segmentation network $F(X : \theta_F)$.

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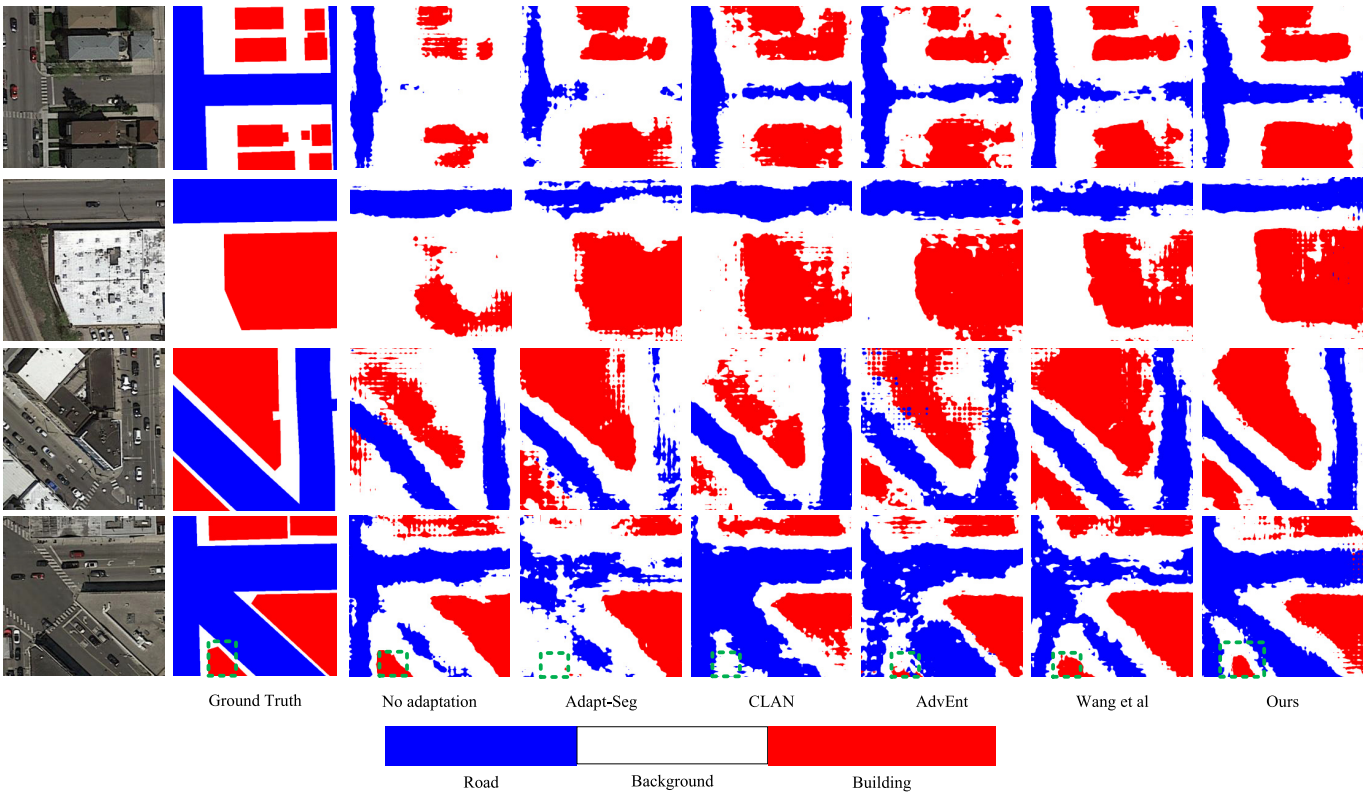



Fig. 6. Visualization of segmentation maps on Chicago test set.

TABLE I

MODEL PERFORMANCE ON TARGET DOMAIN UNDER DIFFERENT SETTINGS ($\varphi_c = 0.95$). WE PRESENT PER-CLASS IOU, MEAN IOU (MIOU), AND OVERALL ACCURACY (OA) FOR EVALUATION

| Method | CCM | ERAM | Imp Surf | Building | Low Veg | Tree | Car | Background/clutter | mIoU | OA |
|---|---|---|---|---|---|---|---|---|---|---|
| No Adaptation | | | 40.96 | 25.93 | 35.57 | 7.46 | 7.86 | 12.9 | 21.78 | 45.8 |
| Baseline | | | 54.33 | 44.00 | 30.77 | 18.45 | **46.49** | 13.16 | 34.54 | 56.3 |
| Baseline | ✓ | | 63.31 | 62.69 | 45.56 | 17.14 | 41.23 | 13.97 | 40.65 | 66.7 |
| Baseline | ✓ | ✓ | **64.76** | **69.92** | **48.2** | **36.68** | 43.04 | **15.47** | **46.35** | **70.6** |

mean IoU (mIoU) of all class to evaluate the overall performance of the model.

### *B. Ablation Experiment*

To verify the effectiveness of the proposed module, we evaluated our model from Vaihingen to Potsdam under different settings and the results are reported in Table I. No adaptation means that we directly use the segmentation model trained with the source domain data to predict the segmentation map of the target domain image. The baseline represents the adversarial training framework in Fig. 2 without CCM and ERAM. And the third row and fourth row display the result of inserting CCM and ERAM on the basis of the baseline gradually. In addition, according to the experimental results, the segmentation model deeplab v2 trained on the source domain can achieve 69.16 of mIoU and 89.3 of OA on the source domain training set, respectively. And it can be seen in first row that mIoU and OA of the no adaptation dropped to 21.78 and 45.8, respectively, which reflects the huge domain gap between the source domain and the target domain. When comparing baseline with no adaptation, it can be significantly found that the adversarial training is able to narrows the gap between domains to a certain extent by aligning the distribution in output space. But at this time, the performance of the model on the target domain is still far from that on the source domain. After inserting CCM to the domain discriminator, the mIoU and OA are further improved to 40.65 and 66.7, which proves it is very effective to transfer the features of each class separately without affecting each other compared with global movement. Finally, compared with just inserting CCM, it can be observed that mIoU increased by 14% and OA increased by 5.6% when we add both CCM and ERAM, which verifies the idea that the features of different regions should be aligned to the source domain with different intensities.

### *C. Comparison With Other Methods*

We compare against other state-of-the-art models that are related to us. To ensure the fairness of comparison, the deeplab v2 with resnet101 is used as the backbone of all models

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TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON POTSDAM TEST SET. PER-CLASS IOU AND MIOU ARE LISTED. THE PERFORMANCE OF ALL MODELS REPORTED IN THE TABLE COMES FROM OUR OWN EVALUATION

| Method | Imp Surf | Building | Low Veg | Tree | Car | Background/clutter | mIoU |
|---|---|---|---|---|---|---|---|
| No Adaptation | 40.96 | 25.93 | 35.57 | 7.46 | 7.86 | 12.9 | 21.78 |
| Adapt-Seg[7] | 59.29 | 56.27 | 43.52 | 9.08 | 46.68 | 11.21 | 37.68 |
| AdvEnt[9] | 60.54 | 56.44 | 44.77 | 11.24 | 47.84 | 12.42 | 38.88 |
| CLAN[10] | 58.42 | 55.71 | 41.8 | 14.22 | 45.33 | 13.16 | 38.11 |
| MaxSquare[12] | 55.12 | 46.62 | 42.53 | 13.82 | 42.93 | 14.75 | 35.96 |
| BDCA[50] | 62.12 | 61.45 | 46.37 | 13.50 | 48.40 | 4.82 | 39.45 |
| Wang et al [18] | 63.58 | 62.61 | 49.73 | 20.30 | 49.67 | 11.36 | 42.88 |
| ColorMapGAN[26] | 64.24 | 59.53 | 49.98 | 18.53 | **53.65** | 12.52 | 43.08 |
| SAC[45] | 65.69 | 67.77 | 49.21 | 37.87 | 47.23 | 14.54 | 47.05 |
| DPL[46] | 66.82 | 68.43 | 48.83 | 36.11 | 49.49 | 16.59 | 47.71 |
| Ours | **67.93** | **72.08** | **51.79** | **39.18** | 50.02 | **16.61** | **49.60** |

TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON CHICAGO TEST SET. PER-CLASS IOU AND MIOU ARE LISTED. THE PERFORMANCE OF ALL MODELS REPORTED IN THE TABLE COMES FROM OUR OWN EVALUATION

| Method | Background | Road | Building | mIoU |
|---|---|---|---|---|
| No adaptation | 45.96 | 16.26 | 39.10 | 33.77 |
| Adapt-Seg[7] | 49.84 | 11.09 | 52.71 | 37.88 |
| AdvEnt[9] | 50.42 | 22.92 | 49.43 | 40.92 |
| CLAN[10] | 53.85 | 23.25 | 44.37 | 40.49 |
| MaxSquare[12] | 53.47 | 20.75 | 43.06 | 39.09 |
| BDCA[50] | 53.88 | 20.31 | 52.97 | 42.39 |
| Wang et al [18] | 53.01 | 23.00 | 55.43 | 43.82 |
| ColorMapGAN[26] | 54.58 | 15.15 | 49.07 | 39.61 |
| SAC[45] | **55.69** | 20.86 | 55.09 | 43.88 |
| DPL[46] | 53.52 | 24.51 | 56.22 | 44.75 |
| Ours | 54.20 | **24.66** | **56.32** | **45.06** |

and all models are well trained. The results are displayed in Tables II and III. The models from the second row to the seventh row are based on adversarial training framework to preform UDA for semantic segmentation of nature images. ColorMapGAN [26] is the UDA model that applies image to image translation framework for semantic segmentation of remote sensing images. Dual path learning (DPL) [46] and self-supervised augmentation consistency (SAC) [45] adopt image to image translation and self-supervised method, respectively. As we can see in Tables II and III, Our model performs better than other models.

In addition to the quantitative comparison, we performed a visual comparison. As illustrated in Figs. 5 and 6, the intuitive segmentation effect of our model is much better than other models and it can be observed that there is no obvious false segmentation in the interior of the object. Apart from this, the red dashed box in Fig. 5 and the green dashed box in Fig. 6 display that our model reduces the phenomenon of negative transfer.

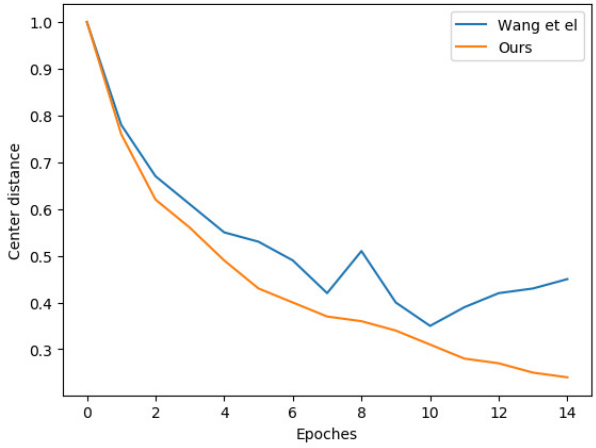



Fig. 7. CCD variation as training goes on.

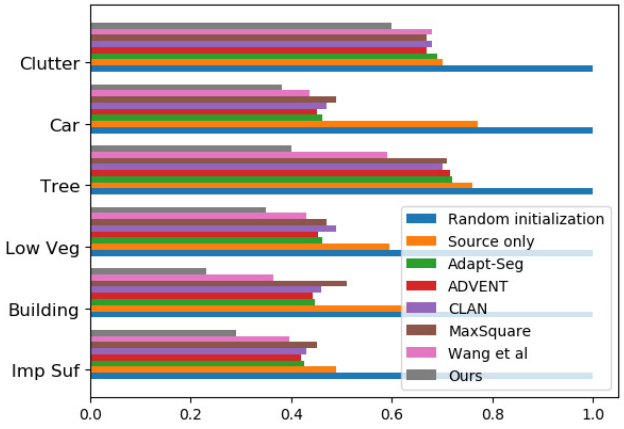



Fig. 8. Quantitative analysis of the feature joint distributions. For each class, we show the distance of the feature cluster centers between the source and target domains.

### *D. Feature Distribution*

According to CLAN [10], to further verify that the ability of our model to reduce the negative transfer phenomenon,

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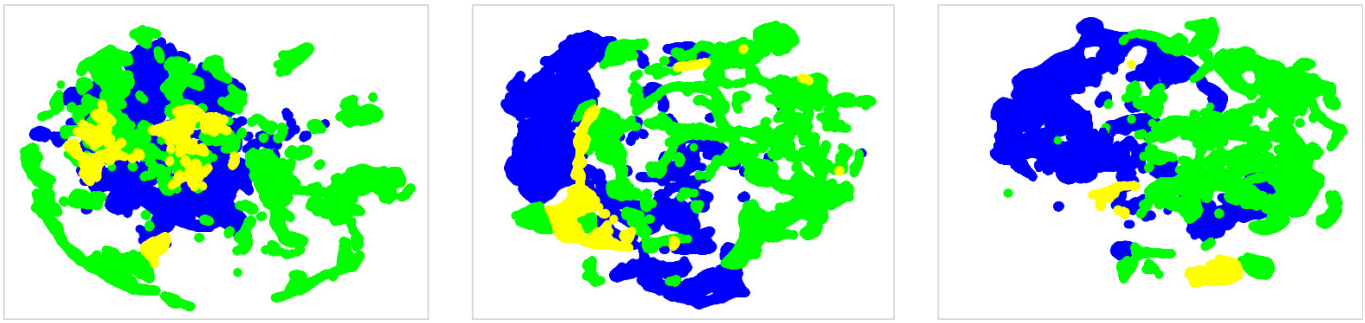



Fig. 9. Comparison of distributions. For a clear illustration, we only show three related classes, i.e., building in blue, tree in green, and car in yellow. Compared with the best adversarial learning framework before us in Table II, the classifier boundary of our model is more obvious, which proves that our model can better align the distribution of each category independently.

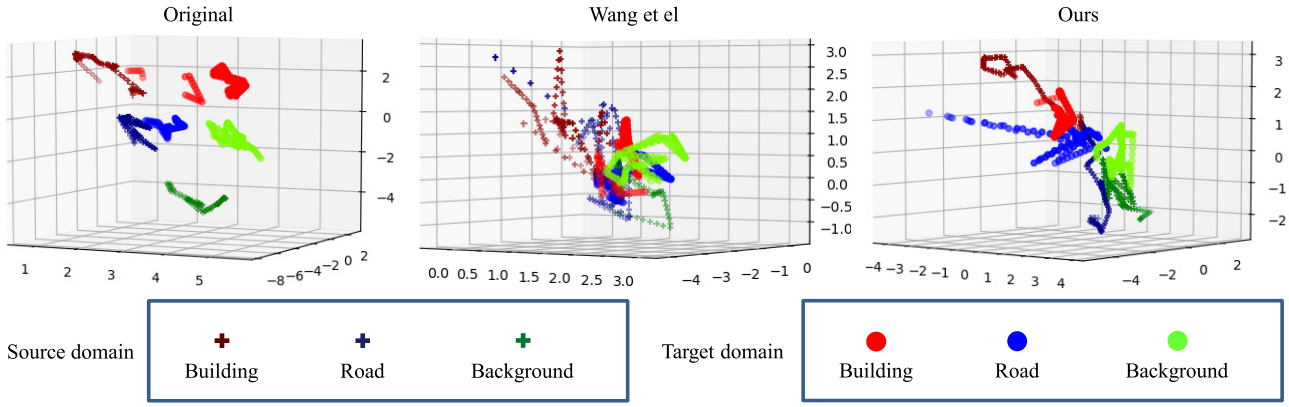



Fig. 10. Comparison of distributions to show the classwise alignment. "Original" represents the feature distribution before adversarial training.

an experiment is designed to delve into the category-level alignment degree of each class. To be specific, we randomly pick out 500 source and 500 target images and then calculate the cluster center distance (CCD) $\{d_1^e, d_2^e, \ldots, d_i^e, \ldots, d_C^e\}$ of the same class's features in output space between the source and target domains, where $e$ and $C$ denotes the number of training epochs and classes, respectively. For better presentation, we normalize $d_i^e$ by $d_i^e/d_i^0$, where $d_i^0$ refers to the CCD from the model whose parameters are initialized randomly. In this way, $d_i^e$ can reflect the relative distance between the source and the target domain in the feature cluster center of the current category. Taking the class "Building" as an example, we display the change of $d_i^e$ with the training epochs increasing as shown in Fig. 7. It can be observed that $d_i^e$ is monotonically decreasing in our model while it is not monotonous in Wang *et al.* [18], which proves our model do not map the well-aligned distribution into error position. Second, Compared with Wang *et al.*, $d_i^e$ converges to a smaller value in our model, which means our model achieves better alignment in output space. The final CCD of each class is shown in Fig. 8. It can be seen that our model can achieve a smaller CCD in most cases compared with other method of adversarial training.

We finally map the high-dimensional features of the segmentation map at last row in Fig. 5 to a 2-D space with T-distributed stochastic neighbor embedding (t-SNE) [43] shown in Fig. 9. The comparison of distributions further proves that our model can enforce category independent alignment during adversarial training. Moreover, we randomly select an image from the Chicago dataset and Paris, respectively, and then display the feature distribution of each class in images before and after adversarial learning to visualize classwise alignment in Fig. 10. It can be seen that before adversarial training, the source and target domain feature distributions are quite different. Although the method of Wang *et al.* [18] aligns the features of the source domain and the target domain globally, the discrimination of each category is still not obvious. More notably, the initial feature distribution of the road is close to alignment before adversarial training. However, some features of road are mapped to the wrong location in the output space after the adversarial training using the method of Wang *et al.* As for our method, it can be observed that the distribution of each category can be aligned separately and the negative transfer is reduced. Note that here we use the method proposed by Wang *et al.* to compare with our method instead of the better-performing ColorMapGAN, SAC and DPL because these three methods are not the method of adversarial training for domain adaptation.

### *E. Pseudo-Label Visualization*

Finding enough and accuracy pseudo label is difficult at the initial training stage when distribution discrepancy between

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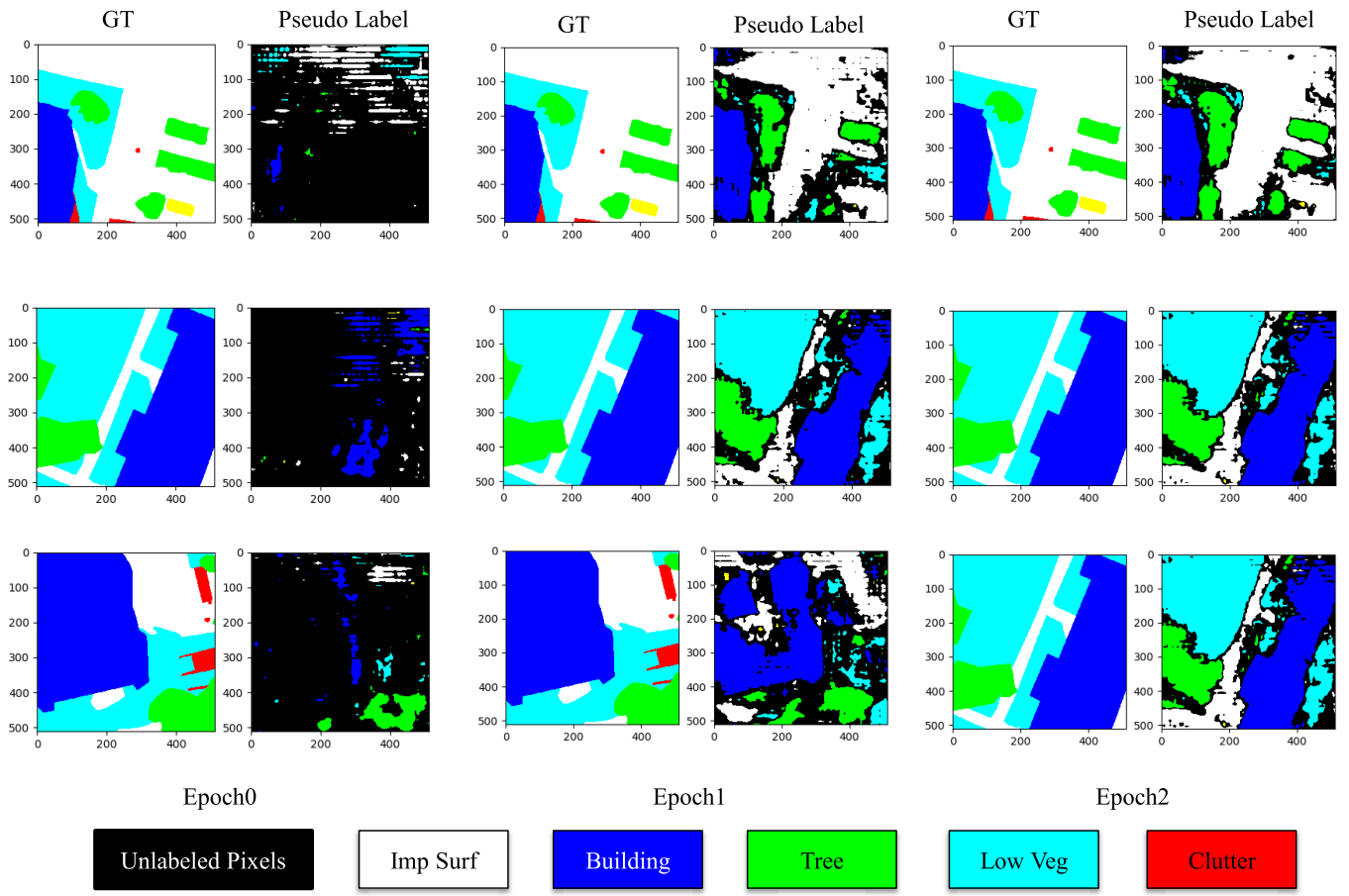



Fig. 11. Change trend of pseudo-label as the training progresses with the setting from Vaihingen to Potsdam when $\varphi_c = 0.80$. The distribution discrepancy between Vaihingen and Potsdam is very large because the model mIoU and OA are 69.16% and 89.3% on Vaihingen dataset and are dropped from to 21.78% and 45.8% when testing on Potsdam dataset without domain adaptation.

two domains is large. To explore the impact of pseudo-labels' quality on classwise adversarial training when domain distributions vary widely, we visualize the changes in pseudo-labels with different $\varphi_c$ as the training process progresses. We find that a relatively big parameter $\varphi_c$ of pseudo label threshold can ensure the accuracy of the obtained pseudo label when the distribution discrepancy between two domains is large. Although the bigger $\varphi_c$ leads to obtain less pseudo labels at initial training stage, the obtained accurate pseudo labels can be used as excellent anchors to promote the classwise alignment at the next stages. In fact, as shown in Fig. 11, the number of pseudo label increases rapidly with the increase of training epochs so that classwise alignment becomes more and more refined even if distribution discrepancy is large. Thus, the proposed method can perform much better than other classwise alignment model like CLAN that is not affected by pseudo-label quality but does not introduce target domain category information at last even through facing a large distribution discrepancy.

## V. Conclusion

In this article, we presented a novel framework to generate high-quality maps from remote sensing images even when there exists a large domain shift between the source and target domains. In our model, we proposed a CCM for aligning the distribution of each class independently and an ERAM for adaptively transferring the distribution of different regions with suitable intensity. We validated our approach on a city pairs, where we performed a city-to-city domain adaptation in them. Our experimental results demonstrated that the proposed framework exhibits a much better performance than the existing approaches.

Furthermore, although the proposal of CCM enables each class to align the distribution in its proper way, the difficulty of aligning the distribution of each class is different due to the same class with different spectrums. Therefore, a possible future research direction is to mine similar prior information like ERAM in CCM so that the model can focus on those classes that are difficult to align. On the other hand, although our method is equally feasible in the UDA of hyperspectral remote sensing images because the increase in the number of input channels will not affect the distribution alignment in the output space, we cannot verify our method on hyperspectral remote sensing images due to the lack of available datasets. Therefore, in future, we will consider creating and releasing the data that can be used for UDA of hyperspectral remote sensing images.

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